

Dquez Carr
MCY 611
Lab 1

MCY 611 Cloud Computing Lab#1: AWS EC2

You will create your AWS account. Then you will use the AWS CLI commands to launch EC2 instances, manipulate security groups, and more.

Tools and Materials

- AWS management console
- AWS CloudShell

3. AWS CLI

4. Ubuntu Server 20.04 LTS AMI: ami-09e67e426f25ce0d7

Note: You need to replace words in *italics* with your information.

Configuring your AWS CLI with your IAM user's access keys

- The AWS CLI comes preinstalled in AWS CloudShell. Let's verify if the CLI is installed properly via the following command:

`aws --version`

What is the output?



```
CloudShell
us-east-1 +
~ $ aws --version
aws-cli/2.23.1 Python/3.12.6 Linux/6.1.119-129.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023
~ $
```

- Configure the AWS CLI with the `aws_cli_user` account (`aws configure`). Use the access key and secret access key values that were generated from the AWS management console. Use the `us-east-1` region as the default region. Use JSON as the default output format.

aws configure

AWS Access Key ID [***F2SP]: AKIA42PHHZCTYBP434RC**

AWS Secret Access Key [***vUSu]: 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**

Default region name [us-east-1]: us-east-1

Default output format [json]: json

Part 3: Creating a security group and a key pair

aws ec2 describe-vpcs --filters Name=is-default,Values=true

What is your VPC ID?

```
~ $ aws --version
aws-cli/2.23.1 Python/3.12.6 Linux/6.1.119-129.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023
~ $ aws configure
AWS Access Key ID [*****34RC]: AKIA42PHHZCTYBP434RC
AWS Secret Access Key [*****aLPU]: qUpIwnwUbF1QlpjDwElenLWTY9nrGQbiemHraLP
U
Default region name [us-east-1]: us-east-1
Default output format [json]: json
~ $ aws ec2 describe-vpcs --filters Name=is-default,Values=true
{
  "Vpcs": [
    {
      "OwnerId": "881490118823",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0688d6164b79f37c6",
          "CidrBlock": "172.31.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": true,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-0e2544332a068204f",
      "State": "available",
      "CidrBlock": "172.31.0.0/16",
      "DhcpOptionsId": "dopt-08b45b45933d7afb0"
    }
  ]
}
```

- Get the subnet ID with the following command:

```
aws ec2 describe-subnets --filters Name=vpc-id,Values=your_vpc_id
Name=availability-zone,Values=us-east-1a Name=default-for-az,Values=true
```

What is your subnet ID?

```

~$ aws ec2 describe-subnets --filters Name=vpc-id,Values=vpc-0e2544332a068204f Name=
AWS CloudShell ty-zone,Values=us-east-1a Name=default-for-az,Values=true
{
  "Subnets": [
    {
      "AvailabilityZoneId": "use1-az6",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "881490118823",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:us-east-1:881490118823:subnet/subnet-0aa6a451bc
da4099f",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-0aa6a451bcda4099f",
      "State": "available",
      "VpcId": "vpc-0e2544332a068204f",
      "CidrBlock": "172.31.32.0/20",
      "AvailableIpAddressCount": 4091,
      "AvailabilityZone": "us-east-1a",
      "DefaultForAz": true,
      "MapPublicIpOnLaunch": true
    }
  ]
}

```

- Create a security group using the VPC ID from the previous command and your NKU username as the group name.

```
aws ec2 create-security-group --group-name your_nku_username --description "my security group"
--vpc-id your_vpc_id
```

What's the output? What is your security group ID?

Details Security Groups			
Security group name  carrd	Security group ID  sg-0dc79f8b11b3a6809	Description  my security group	VPC ID  vpc-0e2544332a068204f 
Owner  881490118823	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

- Add rules to the security group. Define two rules that allow incoming traffic over port 22 and port 80 for SSH and HTTP and add them to your security group via the following commands:

```
aws ec2 authorize-security-group-ingress --group-id your_security_group_id --protocol tcp --port 22
--cidr 0.0.0.0/0
```

```
aws ec2 authorize-security-group-ingress --group-id your_security_group_id --protocol tcp --port 80
--cidr 0.0.0.0/0
```

Filter rules					< 1 >
Name	Security group rule ID	Port range	Protocol	Source	
Instances					
-	sgr-0c3df4c1799e2d078	80	TCP	0.0.0.0/0	
-	sgr-047efaaa5b77a7d69	22	TCP	0.0.0.0/0	

```
aws ec2 describe-security-groups --group-id your_security_group_id
```

```
~$ aws ec2 describe-security-groups --group-id sg-0dc79f8b11b3a6809
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0dc79f8b11b3a6809",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-0e2544332a068204f",
      "SecurityGroupArn": "arn:aws:ec2:us-east-1:881490118823:security-group/sg-0dc79f8b11b3a6809",
      "OwnerId": "881490118823",
      "GroupName": "carrrd",
      "Description": "my security group",
      "IpPermissions": [
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ]
        }
      ]
    }
  ]
}
```

```

        "CidrIp": "0.0.0.0/0"
    },
    "Ipv6Ranges": [],
    "PrefixListIds": []
},
{
    "IpProtocol": "tcp",
    "FromPort": 22,
    "ToPort": 22,
    "UserIdGroupPairs": [],
    "IpRanges": [
        {
            "CidrIp": "0.0.0.0/0"
        }
    ],
    "Ipv6Ranges": [],
    "PrefixListIds": []
}
]

```

- ```
aws ec2 create-key-pair --key-name your_username-key --query 'KeyMaterial' --output text >
devenv-key.pem
```

Did you see devenv-key.pem under your current directory? Is devenv-key.pem your private key or your public key?

You need to modify permissions to the devenv-key.pem file so that only you can access it.

chmod 400 devenv-key.pem

Part 4: Creating an EC2 instance

- To start an instance, you can run the following command:

aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 1 --instance-type t3.micro --key-name *your\_username*-key --security-group-ids *your\_security\_group\_id* --subnet-id *your\_subnet\_id*

What’s the output? What’s your instance ID?

i-0e164d5ddd0c93591

▼ Instance summary [Info](#)

Instance ID

i-0e164d5ddd0c93591

Public IPv4 address

50.17.145.239 | [open address](#)

Private IPv4 addresses

172.31.39.16

IPv6 address

-

Instance state

Running

Public IPv4 DNS

ec2-50-17-145-239.compute-1.amazonaws.com | [open address](#)

Hostname type

IP name: ip-172-31-39-16.ec2.internal

Private IP DNS name (IPv4 only)

ip-172-31-39-16.ec2.internal

Answer private resource DNS name

-

Instance type

t3.micro

Auto-assigned IP address

50.17.145.239 [Public IP]

VPC ID

vpc-0e2544332a068204f

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recom

i-0e164d5ddd0c93591

IAM Role

-

Subnet ID

subnet-0aa6a451bcda4099f

Auto Scaling Group name

-

IMDSv2

Optional

EC2 recommends setting IMDSv2 to required

[Learn more](#)

Instance ARN

arn:aws:ec2:us-east-1:881490118823:instance/i-0e164d5ddd0c93591

Managed

false

Operator

-

▼ Instance details [Info](#)

AMI ID

ami-09e67e426f25ce0d7

Monitoring

disabled

Platform details

Linux/UNIX

AMI name

ubuntu/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-202310430

Allowed image

-

Termination protection

Disabled

- You need the instance’s public IP address or public DNS name to access the instance. After waiting for 20 seconds, you can use the following command to find out the network information:

aws ec2 describe-instances --instance-ids *your\_instance\_id*

What’s the output? What is the public IP address of your instance?

```
us-east-1 +
~ $ aws ec2 describe-instances --instance-ids i-0e164d5ddd0c93591
{
 "Reservations": [
 {
 "ReservationId": "r-00d290ac18f940dea",
 "OwnerId": "881490118823",
 "Groups": [],
 "Instances": [
 {
 "Architecture": "x86_64",
 "BlockDeviceMappings": [
 {
 "DeviceName": "/dev/sda1",
 "Ebs": {
 "AttachTime": "2025-01-27T01:17:06+00:00",
 "DeleteOnTermination": true,
 "Status": "attached",
 "VolumeId": "vol-0a4f39d18d55fee53"
 }
 }
],
 "ClientToken": "a0e4bc3b-6796-42de-afdb-8d56f05b1ecc",
 "EbsOptimized": false,
 "EnaSupport": true,
 "Hypervisor": "xen",
 "NetworkInterfaces": [
 {
 "Association": {
 "IpOwnerId": "amazon",
 "PublicDnsName": "ec2-50-17-145-239.compute-1.amazonaws.com",
 "PublicIp": "50.17.145.239"
 }
 }
]
 }
]
 }
]
}
```

Part 5: logging in to your EC2 instance

- You can use the following ssh command to connect to the instance:  
ssh -i devenv-key.pem ubuntu@*the\_public\_IP\_address\_of\_your\_instance*

```
us-east-1 +
AWS CloudShell
Welcome to Ubuntu 20.04.0 LTS (GNU/Linux 5.4.0-1043-aws x86_64)

* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Mon Jan 27 16:29:01 UTC 2025

System load: 0.24 Processes: 118
Usage of /: 29.2% of 7.69GB Users logged in: 0
Memory usage: 35% IPv4 address for ens5: 172.31.39.16
Swap usage: 0%

* Ubuntu Pro delivers the most comprehensive open source security and
 compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Mon Jan 27 01:24:28 2025 from 44.203.81.102
ubuntu@ip-172-31-39-16:~$
```

If you see “Are you sure you want to continue connecting (yes/no)?”, please type “yes” to continue. When you see the prompt, type “sudo -s” to switch to the root.

```
Last login: Mon Jan 27 01:24:28 2025 fr
ubuntu@ip-172-31-39-16:~$ sudo -s
root@ip-172-31-39-16:/home/ubuntu#
```

- Run the following command on the terminal of your instance:

cat /proc/cpuinfo

What is the output?

```
ubuntu@ip-172-31-39-16:~$ sudo -s
root@ip-172-31-39-16:/home/ubuntu# cat /proc/cpuinfo
processor : 0
vendor_id : GenuineIntel
cpu family : 6
model : 85
model name : Intel(R) Xeon(R) Platinum 8259CL CPU @ 2.50GHz
stepping : 7
microcode : 0x5803801
cpu MHz : 2499.996
cache size : 36608 KB
physical id : 0
siblings : 2
core id : 0
cpu cores : 1
apicid : 0
initial apicid : 0
fpu : yes
fpu_exception : yes
cpuid level : 13
wp : yes
flags : fpu vme de pse tsc mtr pae mca cx8 apic sep mtrr pge mca ovag pat pse36 clflush mmx fxsr sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_
tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave av
x f16c rdrand hypervisor lahf_lm abm 3dnowprefetch invpcid_single pti fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid mpx avx512f avx512dq rdseed adx smap
clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves ida arat pku ospke
bugs : cpu_meltdown spectre_v2 spectre_v2_spec_store_bypass l1tf mds swapgs itlb_multihit
bogomips : 4999.99
clflush size : 64
cache alignment : 64
address sizes : 46 bits physical, 48 bits virtual
power management:
```

- ```
apt update -y
apt full-upgrade -y
apt install apache2 -y
/etc/init.d/apache2 restart
```

```
Processing triggers for dbus (1.12.16-2ubuntu1.1) ...
root@ip-172-31-39-16:/home/ubuntu# apt install apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfswupdplugin1 libxmlb1
Use 'apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0
  ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4
  liblua5.2-0 ssl-cert
0 upgraded, 11 newly installed, 0 to remove and 0 not upgraded.
Need to get 1875 kB of archives.
After this operation, 8121 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 libapr1 amd64 1.6.5-1ubuntu1.1 [91.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1 amd64 1.6.1-4ubuntu2.2 [85.1 kB]
```

- Not Secure — 50.17.145.239

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Lab 1 EC2

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5 Apache2 Ubuntu Default Page: It works


ubuntu

Apache2 Ubuntu Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented** in `/usr/share/doc/apache2/README.Debian.gz`. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/  
|-- apache2.conf  
|   |-- ports.conf  
|-- mods-enabled  
|   |-- *.load  
|   |-- *.conf  
|-- conf-enabled  
|   |-- *.conf  
|-- sites-enabled  
|   |-- *.conf
```

• `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining

- You will be billed as long the instance is running, so you will need to shut it down when you're done. Click + as shown in the figure below to start a new terminal and then click us-east-1.

Use the new terminal to run the following command to terminate your instance:

```
aws ec2 terminate-instances --instance-ids your_instance_id
```

What is the output?

Instance ID i-0e164d5ddd0c93591	Public IPv4 address -	Private IPv4 addresses -
IPv6 address -	Instance state ⊖ Terminated	Public IPv4 DNS -
Hostname type -		
Answer private resource DNS name -	Instance type t3.micro	Elastic IP addresses -

Submission instruction

After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should **write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.**

The goal of this lab is to set up a functional AWS account, gain familiarity with the AWS Command Line Interface (CLI), and use it to manage cloud resources effectively. We concentrated on starting EC2 instances and setting up security groups to manage instance access. You must first establish and configure an AWS account, which includes installing the AWS CLI and configuring the required credentials, in order to accomplish these objectives. After that, you launch EC2 instances using CLI commands, giving important details like the instance type and key pair for secure access. In order to ensure appropriate access control, you lastly modify security groups to permit or prohibit incoming traffic. An essential ability for cloud operations, this practical exercise shows how to programmatically manage AWS resources.